International Journal of Biological and Pharmaceutical Sciences Archive

ISSN: 0799-6616 (Online)

Journal homepage: https://ijbpsa.com/

(Review Article)

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Cannabis sativa: Phytochemistry-An updated Review

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International Journal of Biological and Pharmaceutical Sciences Archive, 2026, 11(01), 020-027

Publication history: Received on 2 December 2025; revised on 15 January 2026; accepted on 20 January 2026

Article DOI: https://doi.org/10.53771/ijbpsa.2026.11.1.0012

Abstract

This review paper highlights latest updates on the phytochemistry of *Cannabis sativa*. *Cannabis sativa* has been used for thousands of years in therapeutic, recreational, and cultural contexts. *Cannabis sativa* contains a peculiar class of phytochemicals known as phytocannabinoids, present almost exclusively in its inflorescences. Phytocannabinoids represent the most important and promising specialized/secondary metabolites from *Cannabis sativa* L. Furthermore, Δ9-Tetrahydrocannabinol (THC) and the corresponding acid tetrahydrocannabinolic acid A (THCA-A), cannabidiol (CBD), cannabinol (CBN), and cannabidiolic acid (CBDA) are the main components of this class. The recently discovered cannabinoid Δ9 -Tetrahydrocannabiphorol (Δ 9 -THCP) was initially known as synthetic derivative, but has been successfully isolated from *Cannabis sativa*, resulting in a product 30 times more active than Δ9 -THC, the main psychotropic compound of Medical *Cannabis sativa* (marijuana). Another newly discovered cannabinoid has been called cannabitriol and can be considered a new CBD derivative. Further cannabinoids have recently been characterized, such as cannabiwinol CBDD, anhydrocannabimovone, isolated for the first time as a natural product, and three new hydroxylated CBD analogues: 1,2-dihydroxycannabidiol, 3,4-dehydro-1,2- dihydroxycannabidiol, and hexocannabitriol, obtained from the crystallization of CBD.

Keywords: *Cannabis Sativa*; Cannabitriol; Cannabiwinol (CBDD); Cannabidiol (CBD); Cannabinol (CBN); Hemp; Phytocannabinoids

1. Introduction

Cannabis sativa L. belongs to the family *Cannabaceae* was used as a medicine before the Christian era in Asia**,** mainly in India**,** China, Bhutan, Nepal, Afghanistan, Pakistan, Egypt, Africa, Iran, Persians and Latin America [1-57]. Medicinal preparations derived from cannabis are rooted in Indian *Ayurvedic* medicine, three main forms are described: "bhang" (prepared from dried leaves), "ganja" (from dried female flowers), and "charas" (resin extracted from the plant's leaves) [1-57]. These traditional forms are not limited to the pharmaceutical sector but also intersect with the food, cosmetics, and recreational industries [1-50]. Being a relatively young and emerging industry with long-term global potential, current agro-industrial transformation studies focus on harnessing cannabis-derived metabolites for the treatment of chronic and neuropathic pain, including fibromyalgia, where *Cannabis sativa* L. is a plant that humankind has been using for millennia [1-57]. The basis of its widespread utilization is its adaptability to so many different climatic conditions, with easy cultivability in numerous diverse environments [1-47]. *Cannabis sativa* has been widely cultivated due to its industrial, ornamental, nutritional, medicinal, and recreational potential [1-57]. From regulatory and application perspectives, cannabis plants are categorized based on the level of Δ9 -tetrahydrocannabinol (THC), one of the most important phytocannabinoids [1-12-50]. *Cannabis sativa* L., is classified into two types as Industrial *Cannabis sativa*,

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(hemp or fibre type) and Medical *Cannabis sativa* L. (drug or marijuana) based on its THC content [1-56]. Medical *Cannabis sativa* (drug or marijuana) contains a very high levels of THC (above 0.3 to 38% of dry weight) [1-40-57]. On the other hand, Industrial *Cannabis sativa* L. (Hemp) contains very low levels of THC (0 to 0.3% of dry weight [1-55]. Plants are generally classified and regulated as industrial hemp if they contain less than 0.3% THC in the dried flower (this level varies by country) or as a drug type with more than this threshold [1-57-68]. The U.S. Farm Bills describe hemp as "*Cannabis sativa* L. and any part of that plant, whether growing or not, with a Λ-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis [1-57]. Hemp is a genetically modified plant with less content of Λ-9 tetrahydrocannabinol (THC) concentration [1-57]. The Industrial *Cannabis Sativa* L. (Hemp) contains very low levels of Λ9-tetrahydrocannabinol (THC) (0 to 0.3% of dry weight) grown outside in a large agriculture farm for the production of fiber, seeds and oil [1-57]. The Industrial *Cannabis Sativa* L. (Hemp) is a short-season crop with many applications which is used as a functional food, fiber, 3D printed building material hempcrete, biochar, bioplastic, paper, textile, biofuel, oil, cosmetics, skincare, animal bedding, ropes, medicine and as a replacement for graphene in supercapacitors [1-57]. Industrial *Cannabis sativa* L. (fiber or Hemp) is legalized by FOOD SAFETY AND STANDARDS AUTHORITY OF INDIA (FSSAI), Government of India, New Delhi since 2021 grown for seed, fiber, medicine, as the functional food, fiber and hempcrete in Uttarakhand state, India [1-57]. The hemp plant will continue to find new applications due to its unique properties for sporting goods, musical instruments, portable and wearable energy storage devices, semiconductors, nanomaterials, active packaging, bio-composite orthotic devices, and medical applications [1-57]. Because of it variegate phytochemistry, *C. sativa* has been used in many sectors, although the discovery of the presence in the plant of several psychotropic substances (e.g., Δ9 -tetrahydrocannabinol, THC) caused a drastic reduction of its cultivation and use together with its official ban from pharmacopeias [1-56]. Fortunately, the discovery of new *Cannabis sativa* varieties with low content of THC as well as the biotechnological development of new clones' rich in many phytochemical components is another milestone in the Cannabis research [1-56]. Concerning its content of secondary metabolites, *Cannabis sativa* has been well studied and as of today about 750 compounds have been reported in this plant [1-56].

The empirical use of the cannabis plant has been practiced for millennia in many cultures to alleviate pain, migraine, and inflammation [1-56]. The medical and therapeutic application of cannabis began to be well understood from the 1960s, when the structure and functional expression of the first G protein-coupled receptor of cannabinoids, named CB1, were characterized in brain tissue [1-65], followed by the GPCR cannabinoid receptor 2 (CB2) which was later identified by homologous cloning in the cells of the immune system [1-65]. In turn, these discoveries led to identification of endogenous CB1 and CB2 ligands, the lipids 2-arachidonoylglycerol (2-AG) and anandamide (the ethanolamide of arachidonic acid) named "endocannabinoids" [1-56]. Cannabinoids—the main bioactive metabolites of *Cannabis sativa*—interact with the human endocannabinoid system (ECS), identified in the early 1990s [1-56-68]. This system regulates essential physiological functions such as cognition, memory, pain perception, sleep, appetite, gastrointestinal processes, and immune response [1-56-68].

2. *Cannabis sativa*: Phytochemistry

Cannabis sativa L. is an ancient crop, but its agricultural adoption has been interrupted to prevent the use of marijuana as a psychoactive drug [1-56-68]. Nevertheless, hemp—the *C. sativa* type with low concentrations of intoxicating Δ9 -tetrahydrocannabinoid—is experiencing a resurgence in interest due to loosened cultivation restrictions and its potential as a multipurpose bio-based crop [1-56-68]. Hemp has valuable applications, including production of medicines from its non-intoxicating cannabinoids, food, medical, and industrial uses of its seed oil rich in polyunsaturated fatty acids, and production of fibers for textiles and industry from its stems [1-56-68]. Recently, several hemp genomic and genetic resources have been developed, allowing significant expansion of our knowledge of major hemp traits, such as synthesis of cannabinoids, oil, and fibers, and regulation of flowering and sex determination [1-56-68]. Still, hemp is an under-improved crop, and its development will depend on the ability to expand and collectively use the novel resources arising from fast advancements in bioinformatics and plant phenotyping [1-56-67]. Cannabis been used for thousands of years in therapeutic, recreational, and cultural contexts [1-56-67]. *Cannabis sativa* contains a peculiar class of phytochemicals known as phytocannabinoids, present almost exclusively in its inflorescences [1-56-67]. Phytocannabinoids represent the most important and promising specialized/secondary metabolites from *C. sativa* L [1-56-67]. The peculiar characteristics of these compounds, as well as their high levels of diverse bioactivities have allowed the production and approval of several cannabis-derived drugs. These substances have also been identified in rhododendron, licorice, and liverwort [12-56-65]. The growing interest in *Cannabis sativa* research has led to the exploration of various techniques for the extraction and analysis of its metabolites. However, despite advances in laboratory methodologies, the industrial application of these processes remains a challenge [12-56-68]. From a chemical point of view, cannabinoids are meroterpenoid compounds with a basic structure of a resorcinyl moiety with different isoprenyl, alkyl, or aralkyl side chains [1-56-65]. Δ9-Tetrahydrocannabinol (THC) and the corresponding acid tetrahydrocannabinolic acid A (THCA-A), cannabidiol (CBD), cannabinol (CBN), and cannabidiolic acid (CBDA) are the

main components of this class. Furthermore, altogether, over 200 different cannabinoids have been isolated from the leaves of *C. sativa*, and the study of the endocannabinoid system and the cannabinoid receptors: CB1 and CB2 has been reported [1-56-65].The study of the endocannabinoid system allowed better understanding of the medicinal effectiveness of cannabis in the treatment of different pathologies[1-56-65]. THCA is the main cannabinoid in the drug-type plant, while CBDA is the main cannabinoid in the fiber-type hemp [1-56-68]. Both acids undergo decarboxylation during storage and in general with heating, giving rise to the phenols THC and CBD, respectively [1-56-65]. While the use and cultivation of drug-type cannabis is subject to legal restrictions in many countries owing to the psychotropic effect of Δ 9 -THC, the hemp type is legal and freely cultivated [1-56-68].

The recently discovered cannabinoid Δ 9 -Tetrahydrocannabiphorol (Δ 9 -THCP) was initially known as synthetic derivative, but has been successfully isolated from *C. sativa*, resulting in a product 30 times more active than Δ 9 -THC, the main psychotropic compound of marijuana [50-65]. More recently, the racemic couple of cis-stereoisomers of Δ 9 -THC were isolated from a low-THC-containing variety of C. sativa registered in Europe as fiber hemp [1-56-65]. Another newly discovered cannabinoid has been called cannabitriol and can be considered a new CBD derivative [30-68]. Further cannabinoids have recently been characterized, such as cannabiwinol CBDD, anhydrocannabimovone, isolated for the first time as a natural product, and three new hydroxylated CBD analogues: 1,2-dihydroxycannabidiol, 3,4-dehydro-1,2- dihydroxycannabidiol, and hexocannabitriol, obtained from the crystallization of CBD [1-56-65]. Following are the list of photocannabinoids have been isolated and identified from plant *Cannabis sativa.*

- Δ 9 -THC δ-9-Tetrahydrocannabinol ($C_{21}H_{30}O_2$) (Mol Wt-314.55).
- Δ 8 -THC δ-8-Tetrahydrocannabinol ($C_{21}H_{30}O_2$) (Mol wt-314.52).
- Δ 9 -THCP (−)-trans-Tetrahydrocannabiphorol ($C_{23}H_{34}O_2$) (Mol wt-342.52).
- THCV Tetrahydrocannabivarin ($C_{19}H_{26}O_2$) (Mol wt-286.45).
- Δ9 -THCA-A δ-9-Tetraidrocannabinolic acid A ($C_{22}H_{30}O4$) (Mol wt-358.47)
- Δ9 b Anhydrocannabimovone $C_{21}H_{28}O_3$ (Mol wt-328.51)
- CBN Cannabinol ($C_{21}H_{26}O_2$) (Mol wt-310.43)
- 7-OH-CBD 7-Hydroxycannabinol ($C_{21}H_{30}O_3$) (Mol wt-330.46)
- CBNA Cannabinolic acid $C_{22}H_{30}O_4$ (Mol wt-358.47)
- CBD Cannabidiol ($C_{21}H_{30}O_2$) (Mol wt-314.46)
- CBDD Cannabidiol dimethyl ether ($C_{23}H_{34}O_2$) (Mol wt-342.52)
- CBD-C2 Cannabidiorcol ($C_{17}H_{22}O_2$) (Mol wt-258.35)
- CBD-DER1 1,2-Dihydroxycannabidiol ($C_{21}H_{33}O_4$) (Mol wt-348.36)
- CBD-DER2 3,4-Dehydro-1,2-dihydroxycannabidin ($C_{21}H_{31}O_3$) (Mol wt-330.37)
- CBD-DER3 Hexocannabitriol ($C_{21}H_{31}O_4$) (Mol wt-348.64)
- CBC Cannabichromene ($C_{21}H_{30}O_2$) (Mol wt- 314.46)
- CBG Cannabigerol ($C_{21}H_{32}O_2$) (Mol wt 316.48)
- CNM Cannabimovone ($C_{21}H_{30}O_4$) (Mol wt- 346.46)
- CBND Cannabinodol ($C_{21}H_{26}O_2$) (Mol wt 310.43)
- CBE Cannabielsoin($C_{21}H_{30}O_4$) (Mol wt 346.46)
- CBF Cannabifuran ($C_{21}H_{26}O_2$) (Mol wt- 310.24)
- CBL Cannabicyclol ($C_{21}H_{30}O_2$) (Mol wt- 314.46)
- CBT Cannabitriol ($C_{21}H_{30}O_3$) (Mol wt-330.46)
- CBR Cannabiripsol ($C_{21}H_{32}O_4$) (Mol wt-348.23)
- CBCA Cannabichromenic acid ($C_{22}H_{30}O_4$) (Mol wt 358.21)
- CBGA Cannabigerolic acid ($C_{22}H_{32}O_4$) (Mol wt- 360.49)
- CBDA Cannabidiolic acid($C_{22}H_{30}O_4$) (Mol wt-358.47)
- CBDP Cannabiphorol ($C_{23}H_{34}O_2$) (Mol wt 342.52)
- CBDD Cannabitwinol ($C_{43}H_{60}O_4$) (Mol wt-640.33)
- CBDV Cannabidivarin (Cannabidivarol) ($C_{19}H_{22}O_2$) (Mol wt- 282.38)
- CBTC Cannabicitran ($C_{21}H_{30}O_2$) (Mol wt- 314.46).

Alkaloids are another important class of secondary metabolites identified in *C. sativa* and isolated from its leaves, stems, pollen, roots, and seeds [1-56-65]. Cannabisativine, anydro-cannabisativine, and the indole alkaloid dihydroxyisoechilunilina have been identified [1-56-65]. Terpenoids represent another important class of secondary metabolites of *C. sativa*. To date, about 200 terpenoids have been detected in *C. sativa*, mainly in leaves, roots, and flowers, as well as in the form of essential oils [1-56-65]. Most of the terpenes isolated from *Cannabis sativa* come from the hydrodistillation of the inflorescences [1-56-65]. Selected mono and sesquiterpenes, together with some non-

terpenoidic compounds and cannabinoids obtained by the hydrodistillation of *Cannabis sativa* inflorescences [1-56-65]. Polyphenols represent a class of specialized metabolites almost ubiquitous within the plant kingdom. Although flavones and flavonols (apigenin, luteolin, kaempferol, and quercetin) are well represented in *C. sativa* aerial parts, this species also contains unique flavonoids such as cannaflavins A, B, and C, all methylated isoprenoid flavones [1-25-65]. Flavonoids constitute about the 10% of the compounds in *Cannabis sativa*, and to date about 26 flavonoids have been recognized [1-65]. Stilbenoids are a further class of phenolic components found in hemp. So far, nineteen compounds belonging to this class have been isolated from *Cannabis sativa*: canniprene, a unique dihydrostilbene, and three new dihydrostilbenoids are reported [1-56-65]. Among the polyphenolic metabolites found in *C. sativa*, a group of peculiar phenolic amides named phenylproprionamides, also known as cannabisins, are worthy of mention. Until now, about 20 cannabisins have been isolated from hempseed [1-56-65]. Biological tests carried out on these particular compounds have revealed their potential as antioxidant and acetylcholinesterase inhibitors, suggesting their suitability in the treatment of Alzheimer's disease [24-56-65].

Phytochemical analyses of roots did not show the presence of cannabinoids, or only in traces, whilst the most important components of this matrix resulted in a series of tri-terpenoids subdivided between triterpenes and phytosterols, some of which are reported [1-65]. The presence of tri-terpenoidic compounds in the roots of *C. sativa* was recently confirmed. According to one of the study the industrial exploitation of the hemp roots of C. sativa as a source of p-coumaric acid derivatives, in consideration of their broad biological activities has been reported [26-65].

Three different types of oil can be obtained from *Cannabis sativa*: (a) vegetable oil, obtained by extraction of seeds and constituted mainly by triglycerides, suitable for human consumption owing to a good balance of ω-6 and ω-3 essential fatty acids (EFAs); (b) essential oil from inflorescences and flowers, obtained by hydrodistillation and steam distillation; (c) so-called 'hashish oil', normally obtained by solvent extraction from plants [1-68]. With the aim to exploit each fraction from the workout of hemp *Cannabis sativa*, seed hulls from *C. sativa* have been also investigated for their content. Using chloroform as extracting solvent and HPLC/MS as an analytical tool, nine diacylglycerols (DAGs), six lysophatidylcholine (LPCs), five lysophosphatidylethanolamines (LPEs), eight phosphatidylethanolamine (Pes), and thirteen phosphatidylethanolamines (PCs) were recovered from the matrix [1-68]. Furthermore, oil from *C. sativa* seeds shows traces of cannabidiol, β-sitosterol, and terpenes together with the tocopherol molecular group including α-, γ-, β-, and δ-tocopherol. The presence of the three main nutrients, namely healthy fat, protein, and fiber provide to the seeds a big opportunity of success, which is also supported by a significant diffusion of vegan/vegetarian diets among the people [1-68]. On the basis of nutritional point of view of *Cannabis sativa* seed oil, it is important to underline that the seeds are particularly rich in omega-3 and stearidonic acids [1-68].

Roots, stems, and leaves from *Cannabis sativa* have recently been investigated to determine their content of tryptophan (TRP) and its metabolic products: kyurenine (KYN) and kynurenic acid (KYNA), which have biological properties [19-65]. Since these compounds are present in higher quantities in the leaves of *Cannabis sativa*, this matrix is proposed as alternative source for these substances in the field of functional foods [1-65]. Further promising therapeutic approaches arise from recent discoveries regarding the efficacy of cannabis-based preparations against intractable epilepsy, brain tumors, Parkinson's disease (PD), Alzheimer's disease (AD), and traumatic brain injury (TBI)/chronic traumatic encephalopathy (CTE). Indeed, extensive observations of seizures in Dravet and Lennox–Gastaut syndrome (LGS) patients indicate good therapeutic success with lower doses of CBD than in Epidiolex® when utilized in cannabis-based preparations with small concomitant amounts of THC, THCA, and linalool, a terpenoid component of cannabis [1-65]. More recently, cannabis preparations with equal THC and CBD content combining THCA and even CBDA along with cytotoxic terpenoids such as limonene have been considered potentially useful in cancer treatment [1-68].

3. Conclusion

Cannabis sativa L. is one of the most versatile plants, with a wide range of heterogeneous applications spanning from the pharmaceutical to the industrial sectors. Although CBD accounts for more than half of the total cannabis market, the progressive liberalization of hemp cultivation and ongoing phytochemical research have sparked growing interest in other cultivars. Both cannabinoids and non-cannabinoid phenolics contribute to the biological activity of *Cannabis sativa* extracts, with cannabinoids playing a central role in antimicrobial responses and stronger anti-inflammatory effect as a pure cannabinoid or as an extract. Phytocannabinoids represent the most important and promising specialized/secondary metabolites from *C. sativa* L. Roots, stems, and leaves from *Cannabis sativa* have recently been investigated to determine their content of tryptophan (TRP) and its metabolic products: kyurenine (KYN) and kynurenic acid (KYNA), which have biological properties. Terpenoids represent another important class of secondary metabolites of *Cannabis sativa*.

Compliance with ethical standards

Disclosure of conflict of interest

No conflict-of-interest to be disclosed.

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